Response to Reviewers

HE-IFD: Privacy-Preserving One-Shot Federated Fine-Tuning under Homomorphic Encryption

Dear Editors and Reviewers,

We thank you for the detailed and constructive reviews. The manuscript has been substantially revised. The principal revision concerns the method, and because it bears on several of the comments, we summarize it before responding to each comment in turn.

Change in method

In the previous version, the encryption-friendly model was trained under homomorphic encryption. That design required polynomial activations in place of ReLU, removed batch normalization, and transmitted large encrypted objects across many internal steps; it was the source of the accuracy loss, the large per-round upload, and the degradation at higher client counts noted in the reviews.

In the revised method, no training is performed under encryption. Each client fine-tunes a small adapter on a frozen, public, pretrained backbone, starting from a shared public initialization, and uploads only the resulting parameter displacement. The server combines these displacements with a single linear operation under multiparty CKKS, after which a threshold of clients decrypts the result. No polynomial activations are used; the encrypted object is the adapter (tens of megabytes rather than hundreds of gigabytes); and the single encrypted operation is exact, so encryption introduces no accuracy cost. This revision addresses comments R1-W1, R1-W2, R1-W3, R2-Q3, and R2-Q6, as detailed below.

Reviewer 1

- **W1 (degradation at higher client counts).** This arose from training from scratch under fragmented data. With a frozen pretrained backbone the model no longer degrades: accuracy is stable from 10 to 100 clients. The residual gap is a function of label heterogeneity rather than client count, and the revision characterizes it explicitly: it is negligible when client data are close to uniform and grows only under severe skew.

- **W2 (operator replacement and its unquantified cost).** Operator replacement no longer occurs. The backbone is evaluated in plaintext on each client, and only a linear sum is encrypted; this encrypted sum reproduces the plaintext result to a relative error of about 10^{-9}. The cost of the cryptographic protocol is therefore isolated and shown to be zero.

- **W3 (upload size and bandwidth).** The per-client upload is now approximately 19 MiB, calculated for a rank-8 adapter, in a single round. A dedicated communication analysis, with a figure and a comparison table, has been added.

- **W4 (empirical privacy; leakage from the released model).** A formal guarantee is now provided for the client contributions: a simulator argument (Proposition 1) establishes that the server's view is computationally independent of the private data under standard CKKS assumptions. The only quantity revealed in plaintext is the final shared model, which any such protocol must reveal; its residual leakage is measured with membership inference.

- **W5 (no NLP; missing ablations).** The evaluation now centers on language tasks (AG-News, TREC, DBpedia, Banking77) and includes a vision task (CIFAR-100). Ablations cover the trainable unit (linear head versus adapter), label heterogeneity, client count, and trajectory length.

Reviewer 2

- **Q1 (comparison with prior work).** A comparison table has been added, positioning the method against representative one-shot distillation, differentially private one-shot, and encrypted federated-learning methods along the axes of one-shot operation, privacy type, number of rounds, reported accuracy, and principal limitation.

- **Q2 (end-to-end CKKS measurements).** The full protocol is implemented end to end in multiparty CKKS using the Lattigo library (key generation, encryption, aggregation, and threshold decryption), and per-operation timings and communication are reported. As the only encrypted operation is a linear sum, the circuit requires no rotations, no relinearization, and no bootstrapping; server memory is approximately 38 MiB and is independent of the number of clients; and, as there is no encrypted training loop, no convergence behavior under encryption arises. Correctness is verified to a relative error of about 10^{-9}.

- **Q3 (client-side encryption cost).** Client-side encryption is approximately 84 ms per client for the adapter, and the upload is 19 MiB; both are derived from measured per-operation rates and reported in the cost section.

- **Q4 (malicious or colluding clients).** A future-work paragraph has been added. It states the threat under an honest-but-curious assumption and identifies two defenses compatible with encryption: robust aggregation evaluated homomorphically (a coordinate-wise median or trimmed mean), and an upload-time norm bound enforced by a zero-knowledge proof.

- **Q5 (post-release privacy across N and α).** Membership inference against the released model, the only plaintext channel, is reported; leakage remains near chance on the language task.

- **Q6 (plaintext weights, ciphertext-by-plaintext products, no encrypted optimizer states).** This corresponds to the present design. The weights and the initialization are public plaintext scalars, only the displacements are encrypted, the server performs plaintext-by-ciphertext multiplication and ciphertext addition exclusively, maintains no encrypted optimizer state, and operates at multiplicative depth one.

Reviewer 3

- **1 (participation incentive).** The incentive is now stated explicitly: each client obtains a model that classifies the entire label space, including classes absent from its own data, which no client could obtain independently.

- **2 (challenges and contributions).** The exposition has been restructured around a single difficulty (a linear average of independently trained models fails under heterogeneity), a single solution (a frozen pretrained backbone supplies the shared frame and removes the alignment phase), and contributions that map directly onto them.

- **3 (motivation placement).** The motivation for one-shot communication and for encryption now appears in the introduction.

- **4 (overview figure).** The protocol figure has been redrawn to show the frozen backbone, the shared initialization, the encrypted displacements, the depth-one server aggregation, and threshold decryption.

- **5 (limited comparison; method not explained).** The comparison now includes FedDF-, FedMD-, and FedAvg-class methods and encrypted baselines, and the experimental section explains the method's operation rather than only reporting outcomes.

- **6 (future directions).** Future directions are now presented in a scope-and-limitations discussion, separate from the cost analysis.

Presentation and readability

The manuscript has been rewritten for clarity, with plain language, consistent notation, a single notation table, and a uniform structure that states the problem, the idea that addresses it, and the supporting evidence in that order. Unsupported qualitative claims have been removed, and each remaining claim is tied to a measurement or a formal argument.

We hope these revisions address the concerns raised, and we thank the reviewers for feedback that has materially improved the paper.